Critical values for Rasch item fit statistics

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Item fit statistics are arguably the most relevant and certainly the most commonly reported feature in Rasch validation studies. A very wide range of item fit statistics have been developed, but very few are used in practice. The most widely used are the infit and outfit mean square item fit statistics (Wright & Stone, 1979; Wright & Masters, 1982) that are in Winsteps (Linacre, 2011) and the fit residual, Chi-square and ANOVA item fit statistics in RUMM2030 (Andrich, Sheridan, & Luo, 2010). All of these are based on residuals, and furthermore they rely on assumptions about convergence to asymptotic distribution of the sum of squared Pearson-type standardized residuals. However, this assumption is problematic (Jennings, 1986). We study the behavior of the most widely used item fit statistics too see when they produce empirical Type I error rates that are close to the nominal 5%. This leads us to recommendations regarding reasonable critical values. We do so in a way that reflects the way most users approach evaluation of item fit, i.e. looking at the evidence for all items and then evaluating if the item that has the worst fit is a cause for concern when taking the total number of items into account. Thus, we provide guidelines for evaluation of item fit that addresses the issue of multiple testing.

References

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